CEITEC

MUNI



C9555: A practical guide to simulations

WEEK 9

Denys Biriukov – Nov. 13th 2025

Get ready for practical session

- Open new terminal `Ctrl + Alt + T`
- Course materials at: https://gitlab.com/denysbiriukov/survival_guide_simulations_c9555
- Any technical issues/problems/questions? Checklist:
  - Access to Wolf.
  - Proper infinity and uv environments.

More a discussion than monologue

If you have a quick question to catch up, raise your hand and ask – we will try to fix it and bring you back

If quick answer does not help, let's chat during a break or after

This course is heavily practical – theory will be trying to catch up, not vice versa

Get ready for practical session

- A bit of things to be done:
  - `mkdir C9555`
  - `mv ~/survival_guide_simulations_c9555/week07/ ~/C9555/`
  - `rm -rf {SURVIVAL_GUIDE_DIRECTORY}`
    - **BE SURE THAT PREVIOUS COMMAND (MV) WAS SUCCESSFUL BEFORE DELETING**

- https://gitlab.com/denysbiriukov/survival_guide_simulations_c9555

Get ready for practical session

- `git clone --depth 1 https://gitlab.com/denysbiriukov/survival_guide_simulations_c9555.git`
- `cd survival_guide_simulations_c9555/`
- `cd week09/`

- `bash ../infinity/infinity-temp.sh`
- `bash ../uv/uv-activate.sh`
- `bash ../uv/jupyter-start.sh`

What is today?

- Week 9:
  Multicomponent systems
- Moving beyond single-phase, homogeneous systems.
- Building complex assemblies: practical know-how.
- Reproducing realistic experimental and relevant conditions.

TEACHING WEEK	DATE	LECTURER	TOPIC
01	18.09	WM	Simple MD to Describe Materials Part I
02	25.09	WM	Simple MD to Describe Materials Part II
03	02.10	IK	Visualization
04	09.10	DB	How to run MD simulations?
05	16.10		**BREAK**
06	23.10	IK	Sampling in Molecular Dynamics
07	30.10	DB	Adding salt to our water
08	06.11	DB	Membranes in MD & CHARMM-GUI
09	13.11	DB	Multicomponent systems
10	20.11	IK	
11	27.11	IK	
12	04.12	WM	
13	11.12	WM	

Today program

Lecture

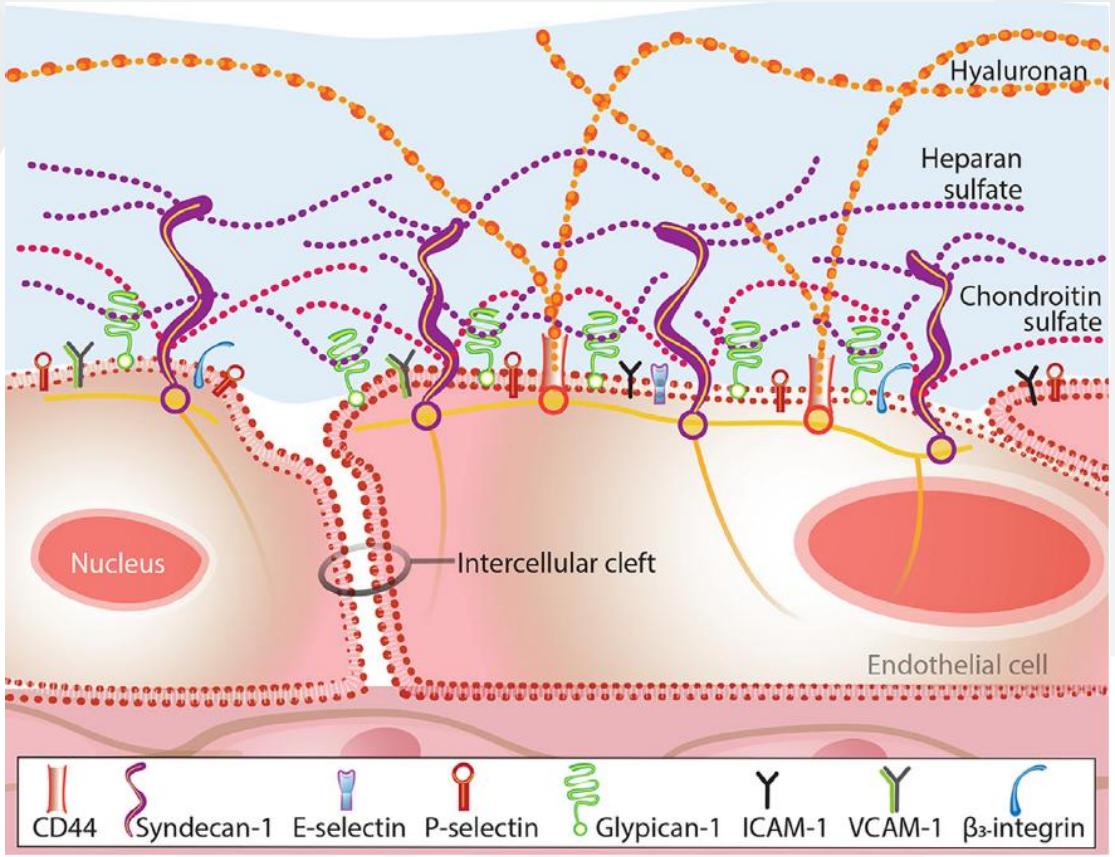
- Building systems beyond water and ions.
- How to make multicomponent systems realistic and working.

Practical session

- Stepwise assembly of peptide-membrane-solvent system.
- Positional restraints.
- Tips, validation checks.

Why do we need complexity?

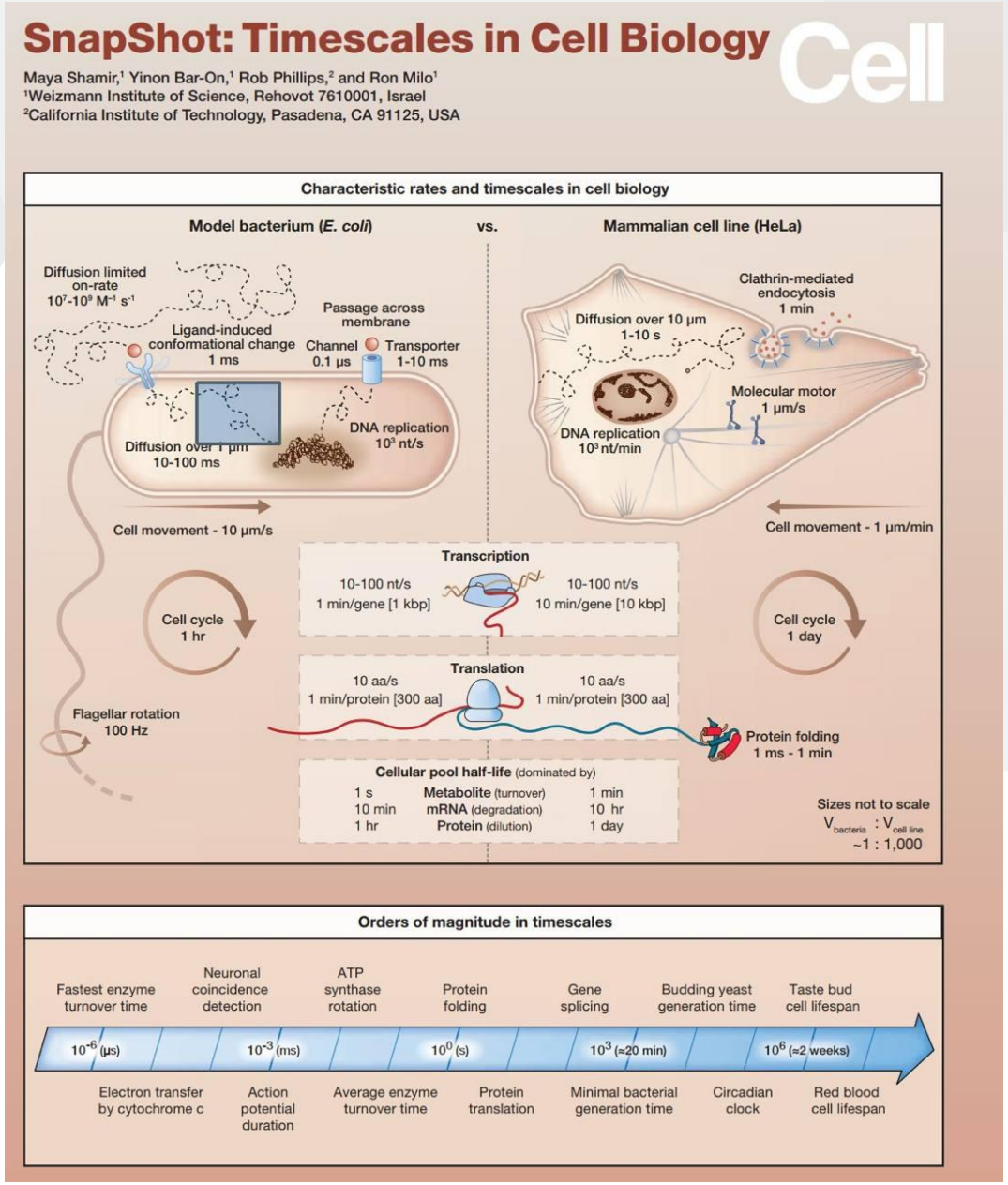
- Real (biomolecular) systems are rarely uniform.
- Simple solvent boxes might be not enough.
- Multicomponent systems provide sufficient context → closer to reality.
- Increased complexity → increased need for control and validation.



[Gaudette, S. et al. J Vet Emerg Crit Care. 2020, 1–18]

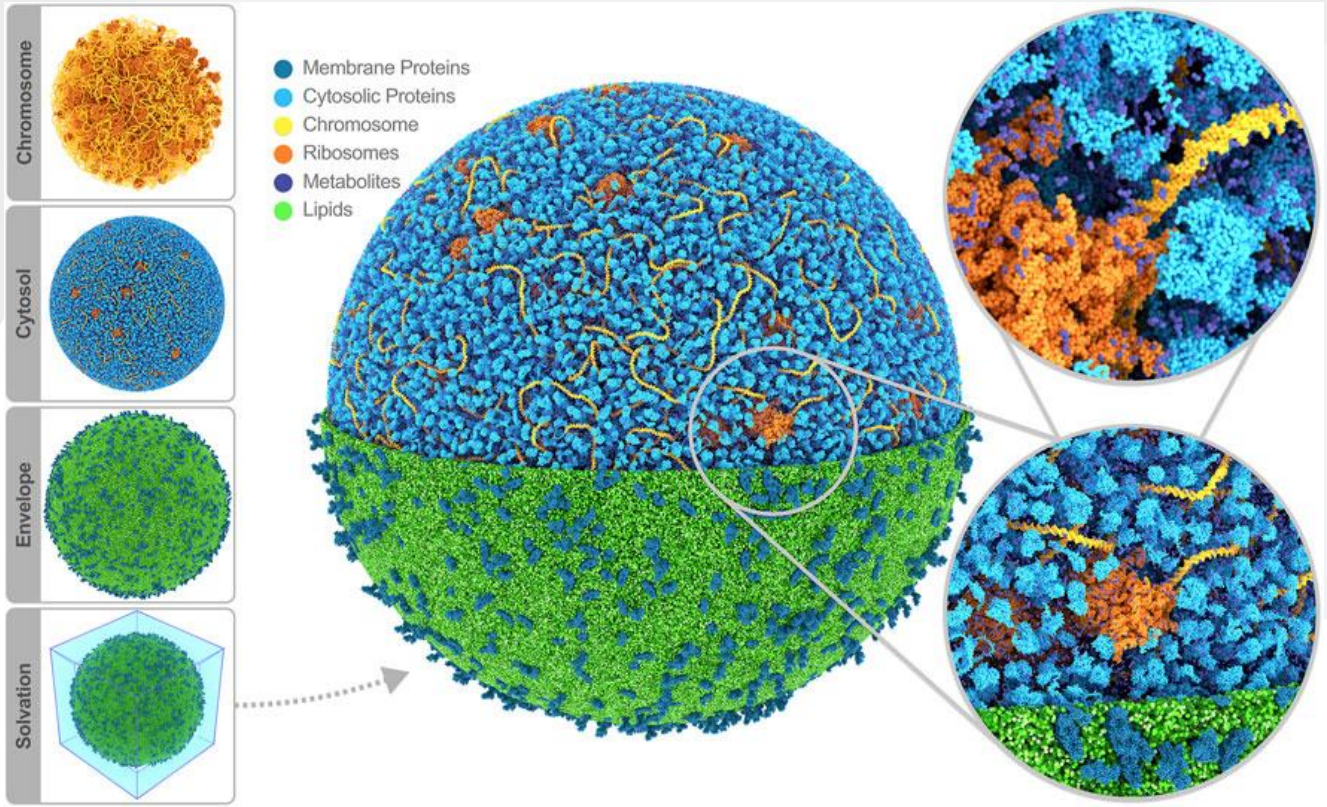
Main challenges

- Different components evolve on distinct timescales.
- Proper mixing and equilibration are more difficult to achieve.
- Interfaces and boundaries require careful setup and equilibration.
- Minor setup inaccuracies can amplify during simulation.



Realism vs. pragmatism

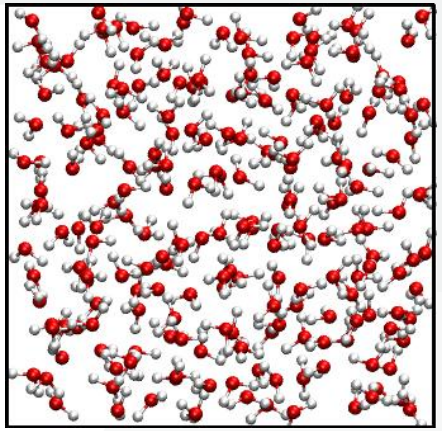
- You cannot run a simulation of an entire body or even a cell (in progress though).
- Keep the model as simple as possible, but no simpler.
- Include only those components essential to the scientific question.
- Balance realism with computational efficiency.

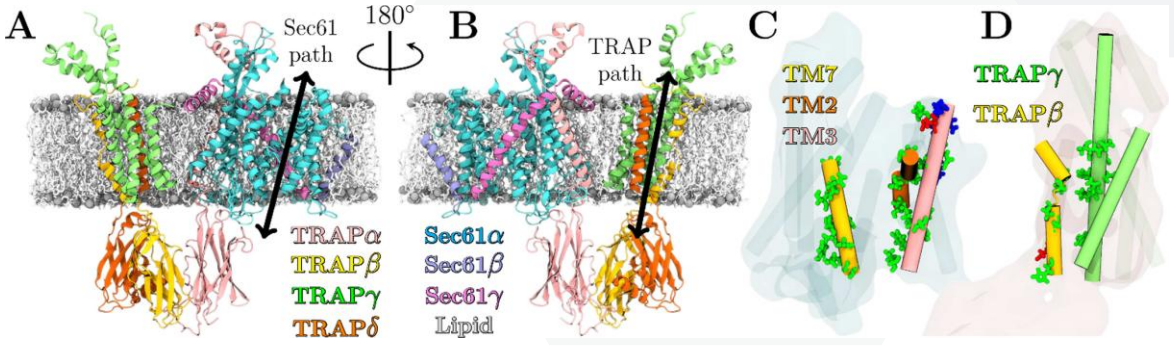


[Stevens, J. A. et al. Front. Chem. 2023, 11, 1106495]

Choosing the starting point

- Decide: build from random mix or predefined structure.
- Some systems can self-assemble within simulation time.
- Others must start from an (experimentally) known state.
- Focus on what can realistically evolve during your simulation run.

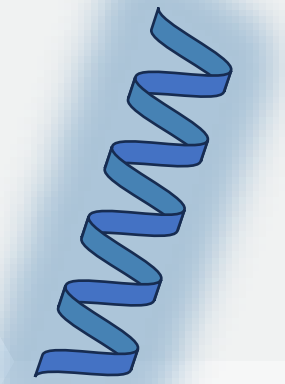


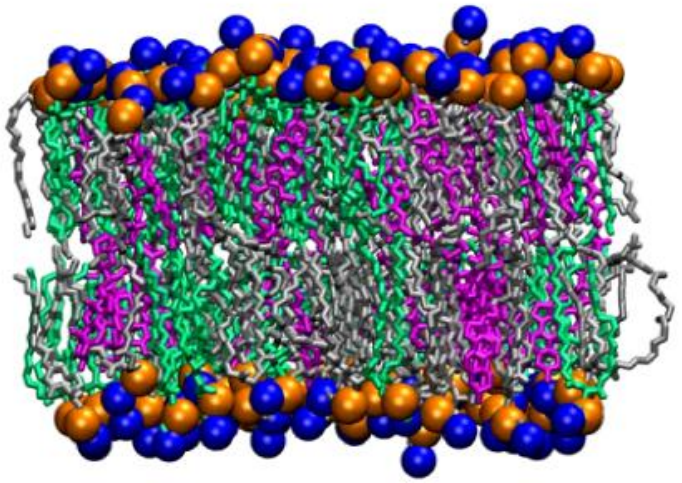


[Javanainen, M. et al. J. Am. Chem. Soc. 2025, 147, 19, 15970–15984]

Choosing the starting point: examples

- Transmembrane peptide:
  - inserted across the bilayer
- Peripheral peptide:
  - placed on the surface, partially embedded
- Cell-penetrating peptide:
  - placed away from the membrane/near the surface/inside the membrane
- Peptide with unknown properties/function:
  - depends on your research question

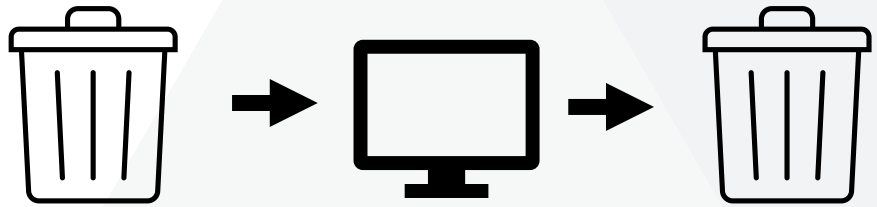




Building a multicomponent system

- Start with the most complex or static components.
  - Add large, rigid structures first; small, flexible ones later.

- Verify that major components are realistic and "**correct**".
  - "Garbage In – Garbage Out".

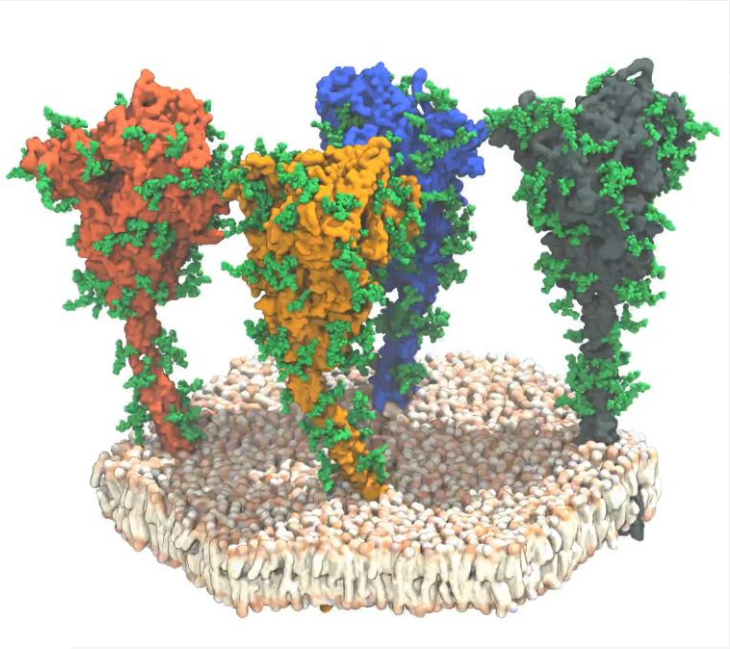


- Proteins: missing residues/atoms, broken bonds, protonation states, termini.

- Membranes/surfaces: orientation, tilting, gaps, or overlaps.

Assembling the components

- Align all components in a common coordinate frame/system before merging.
- Check orientations of all components.
- Remove any steric clashes or unrealistic overlaps early.
- Define key reference axes (*e.g.*, membrane normal, surface normal, ionic channel).

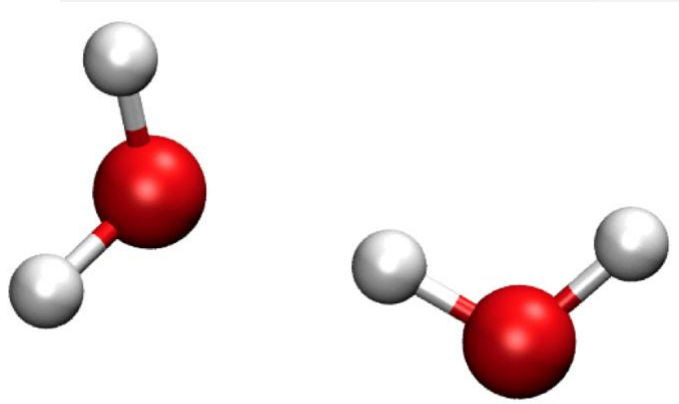


Spike protein of SARS-CoV-2

TUROŇOVÁ, SIKORA, SCHÜRMANN et al.
MAX PLANCK INSTITUTE OF BIOPHYSICS
PAUL-EHRLICH-INSTITUT

Solvating the system

- Done at later stages.
- Ensure full solvation (hydration) of all relevant components/their parts.
- Avoid vacuum gaps and uneven solvent layers.
- Verify solvent distribution visually.
- Respect chemistry – water placement must make physical sense (if you are lucky, the simulation may correct it by itself).



Ion/salt placement and charge balance

- Neutralize the system before production runs.
- Add ions to match experimental or physiological or relevant salt concentration.
- Place ions away from charged groups to avoid possible artifacts.
- Check that ions are evenly distributed, not clustered.
- Know your system: ion type and concentration can impact things a lot.



www.freepik.com

System assembly tools

- If an automatic builder handles it end-to-end, great, you are lucky!
- However: trust but verify.
- Manually inspect coordinates and topology consistency.
- Verify atom naming, periodicity, box size, unrealistic overlaps, …
  - Visualization often helps a lot!

CHARMM-GUI

Effective Simulation Input Generator and More



PACKMOL

Force field consistency

- Use a single force field family for all components.
- Avoid mixing parameters from different sources or versions.
- Ensure each component has valid (and distinct) atom types.
- Check water and ion models are compatible with the chosen force field.
- Re-parameterize or validate any custom molecules before inclusion.

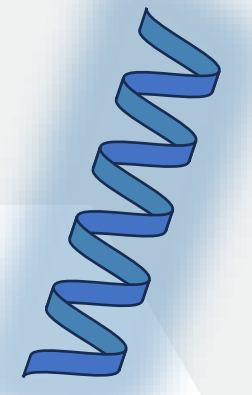
CHARMM
Chemistry at HARvard Macromolecular Mechanics

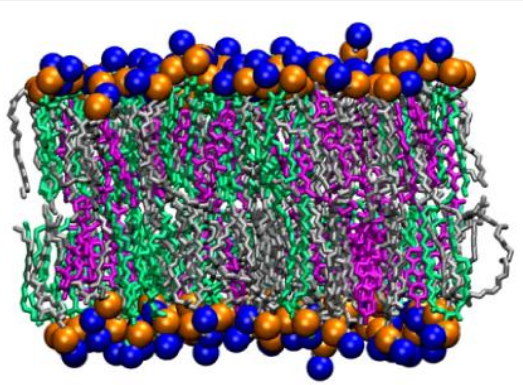


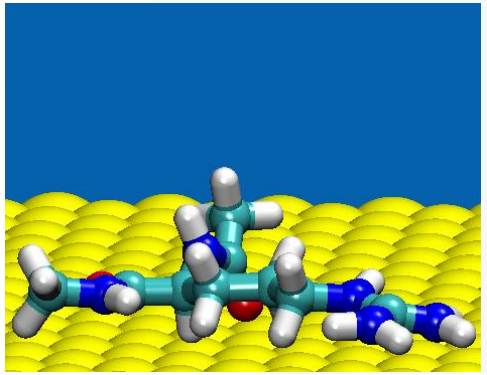
The Amber Project
Chemistry through a Computational Lens

“Maintain” the reality

- Can your protein unfold?
  - Maintain secondary structure when needed.
- Can your material deform?
  - Preserve realistic geometry.
- Can your membrane porate?
  - If not intentional, something is wrong.
- Can the molecule detach from/escape the membrane/surface?
  - Maybe it should (soluble peptide), maybe not (transmembrane peptide).

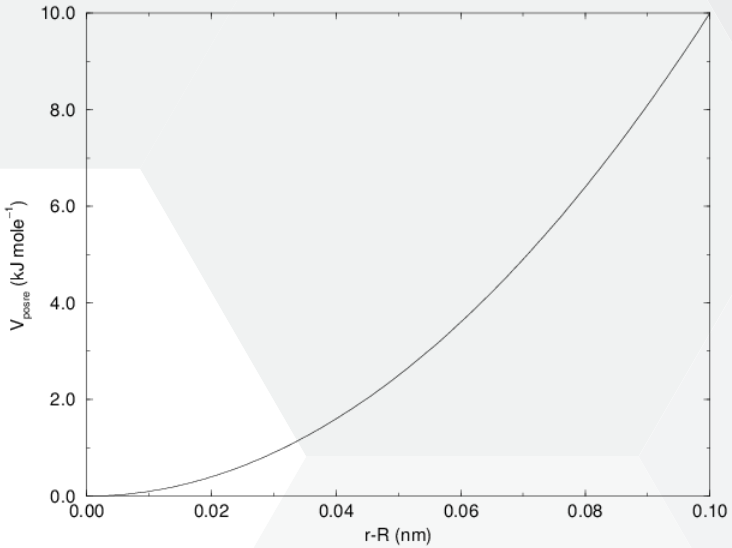


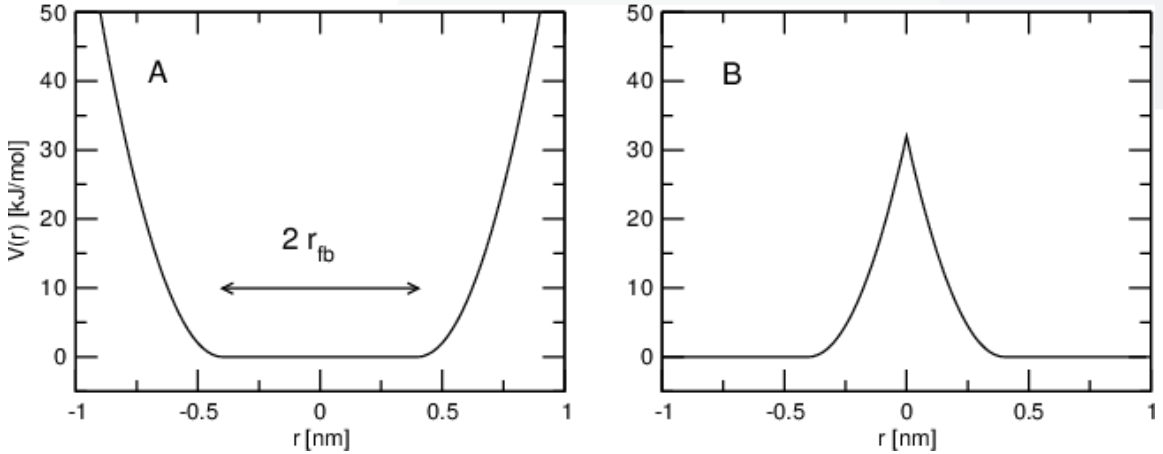




Restraints and constraints

- Restraints: allow motion but keep atoms near a reference position or distance
  - Positional, dihedral, flat-bottom, …
- Constraints: fix certain degrees of freedom completely (*e.g.*, bond lengths)
- Common uses: maintaining geometry, stabilizing interfaces, or holding reference atoms
- Essential during equilibration; optional or selective during production runs

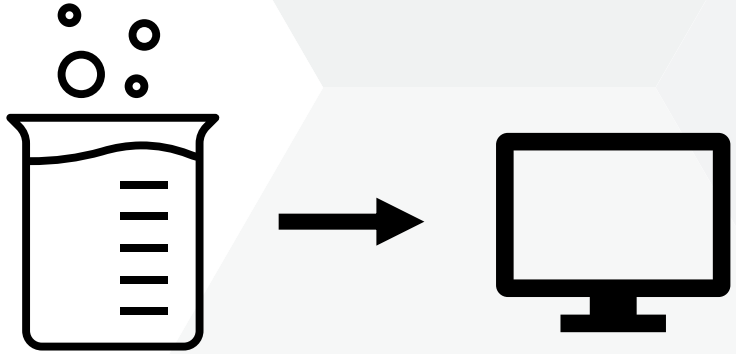




https://manual.gromacs.org/

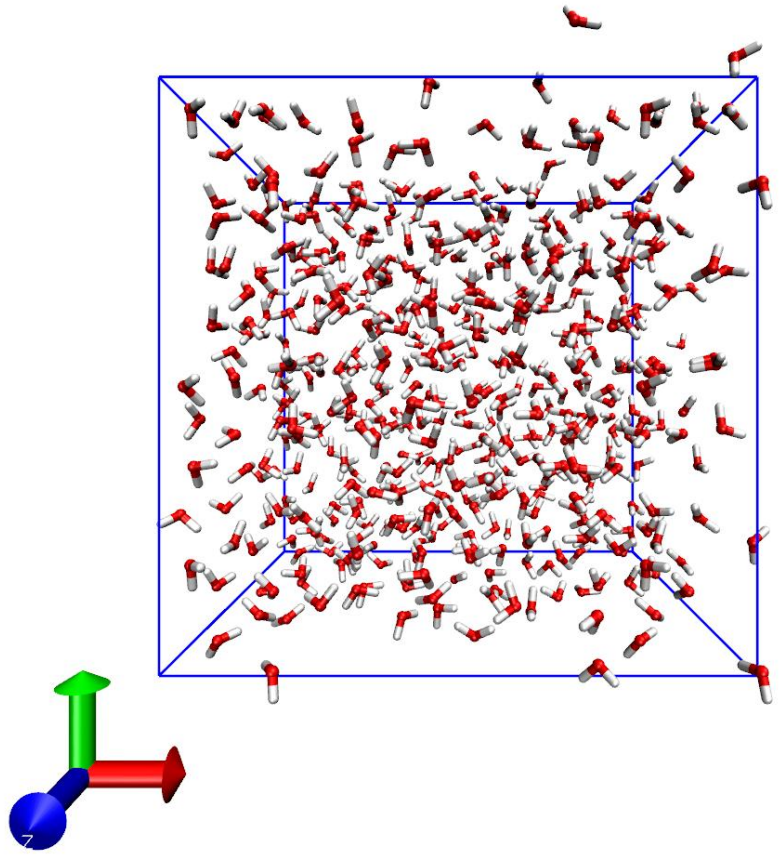
Matching experimental/relevant conditions

- Match temperature, pressure, ionic strength, … to experimental or/and relevant settings.
- Adjust pH through protonation states or constant-pH methods.
- Include ions, buffers, or crowding agents when needed.
  - They are often not central to the main question, but still important.
- Remain pragmatic.



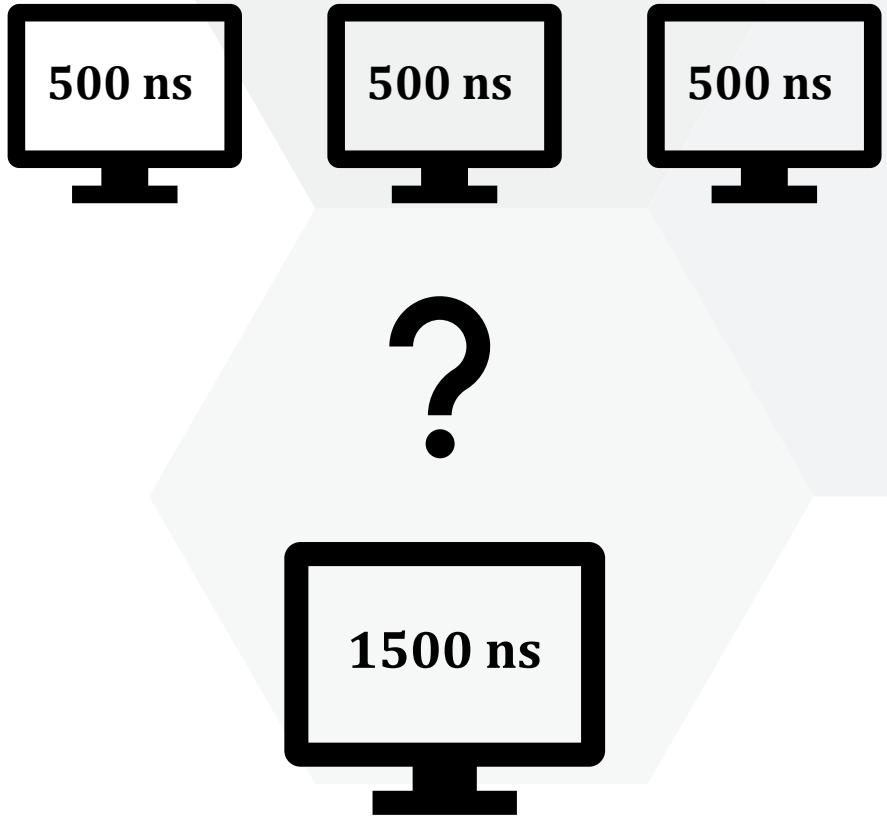
Equilibration protocol

- Begin with energy minimization to remove clashes and other possible instabilities.
- Run short *restrained* simulations (NVT, NPT) to relax the system.
  - Equilibration time varies.
- Gradually release restraints on all components.
- Monitor energy, temperature, pressure, density, … convergence before moving to production stage.



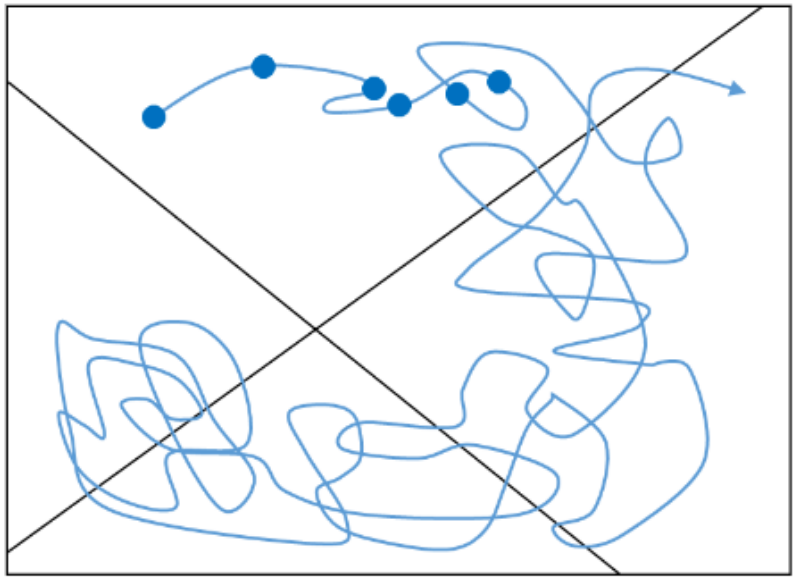
Simulation replicas and timescale

- Match timescale to the process of interest.
- Replicas improve sampling and reproducibility.
  - The same mindset as with experiments.
  - Replicas better than one longer simulation.
- Greater complexity demands longer timescales (big ≠ complex).
- Remain pragmatic with available resources.
  - All-atom: 1–5 μs
  - Coarse-grained: 10–100 μs



Convergence and reproducibility

- Verify stability of key observables (*e.g.*, RMSD, area per lipid, secondary structure, …).
- Compare replicas for convergence rather than focusing on trajectory length.
- Identify decorrelation time before production analysis.
- Document random seeds, parameters, and protocols for reproducibility.



[Grossfield, A. et al. Living J. Comp. Mol. Sci. 2019, 1(1), 5067]

Best Practices for Quantification of Uncertainty and Sampling Quality in Molecular Simulations [Article v1.0]

Alan Grossfield[1*†], Paul N. Patrone[2*†], Daniel R. Roe[3*†], Andrew J. Schultz[4*†], Daniel W. Siderius[5*†], Daniel M. Zuckerman[6*†]

Break?

Get ready for practical session!